PALATIAL POOL SERVICES
Proprietary Technology

ESP32 Diagnostic Tools

Precision IoT Sensor Arrays for Complex Diagnostics

Our custom-engineered IoT diagnostic arrays deploy temporarily to investigate complex pool issues that standard equipment cannot detect. These sensor suites monitor water chemistry micro-fluctuations, electrical current draw, and fluid dynamics at high frequency — identifying intermittent faults, stray voltage, and chemical imbalances with laboratory-grade precision.

Key Benefits

- Detects intermittent faults invisible to standard equipment
- Identifies dangerous stray voltage in pool water
- Laboratory-grade chemical analysis without lab delays
- Non-invasive installation — no plumbing or electrical modifications
- 48-hour battery backup ensures uninterrupted monitoring

At a Glance

Processor: Dual-core ESP32 at 240 MHz with 520KB SRAM

Sensors: pH, ORP, TDS, current transformers, pressure transducers, flow meters

Sampling Rate: 10Hz telemetry, aggregated into 1-second payloads

Connectivity: Wi-Fi 802.11 b/g/n, Bluetooth 4.2, LTE-M/NB-IoT fallback

Battery: 18650 lithium-ion, up to 48 hours backup

Deployment: IP67-rated Pelican case, 3-7 day monitoring periods

Ready to upgrade your pool service experience?

Call **0452 157 694** or visit **beta.palatialpools.com.au**

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